Bisoj Biswakarma

Frontend Engineer · React, Next.js, TypeScript

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Summary

Frontend engineer with 3+ years building production React and Next.js products — currently remote with a US insurance company as architect of the Next.js monorepo behind a marketplace used by **3,000+ agents**; previously **public-sector platforms** including Nepal's national census portal. Strong in performance, accessibility (WCAG 2.2), design systems and developer experience; owns features end to end, from API and Salesforce integration to analytics. Maintainer of **a11y-beast**, an open-source WCAG 2.2 scanner.

Experience

Software Development Engineer, *Renegade Insurance — Remote (US)* *Apr 2025 – Present*

- Architected **AgencyHeight**, a Next.js/React insurance marketplace used by 3,000+ agents, from scratch as a Turborepo + pnpm monorepo with 10+ shared packages (ui, auth, api, state, analytics).
- Cut **bundle size by 30%** and improved Core Web Vitals scores by **46%** versus the legacy platform.
- Set the team's frontend standards, now followed across the codebase: strict TypeScript, Biome, Husky + commitlint, a feature → dev → qa → production flow and Dockerized builds.
- Review pull requests, estimate work and prepare releases through that flow, investigate production incidents and maintain the team's technical documentation.
- Built the design system (shadcn/ui, Radix UI, shared Tailwind preset, Ladle stories) and the data layer (TanStack Query, Zustand, React Hook Form + Zod, Axios with auth/error interceptors).
- Set up Vitest (unit), Playwright (E2E) and axe-core testing; wired Sentry and Datadog for observability.
- Led the insured customer portal (React + TypeScript), now driving **70–100 policy inquiries a week at about 20% conversion** into the agent sales pipeline; shipped onboarding and refactored legacy screens.
- Virtualized a carrier-selection dropdown of thousands of items to remove lag in agent workflows; integrated Google Places API for agent reviews; remediated OWASP ZAP findings (CSP, clickjacking, SRI, CORS).
- Contributed LWC/Apex changes and integration work to the Salesforce Agent Portal, bridging agent tooling with the Next.js stack.

Software Engineer, *Young Innovations — Nepal* *Aug 2023 – Apr 2025*

- Built the interactive visualization layer of the **Census Nepal 2021** portal for the Central Bureau of Statistics, turning raw demographic data into accessible, citizen-facing charts on tight government timelines.
- Shipped internal admin dashboards for the **Municipal Association of Nepal** (293 municipalities) and the World Bank–funded **DGM Nepal** program: data visualization, user management and daily reporting.
- Owned the frontend of **Sarang**, a responsive travel-guide platform with interactive itineraries, built for low-bandwidth mobile users and desktops across Nepal.
- Collaborated with product, design and backend teams to ship filtering, analytics dashboards and dynamic forms against REST APIs.

Frontend Developer, *Brandspire Creatives — Nepal* *Apr 2023 – Aug 2023*

- Built conversion-focused marketing sites and custom WordPress solutions for construction and consultancy clients, translating designs into pixel-accurate, responsive front-end builds.

Full-Stack Developer Intern, *Hyperce — Nepal* *Jan 2023 – Mar 2023*

- Shipped frontend features, user-facing optimizations and production bug fixes in a fast-paced startup.

Projects

Commercial

AgencyHeight — Insurance marketplace for 3,000+ agents *Renegade Insurance · internal*

- Architected from scratch as a Next.js/React Turborepo monorepo with 10+ shared packages (ui component library, auth, api, state, analytics); 30% smaller bundle than the legacy platform.

Renegade Customer Portal — Insured-facing policy portal *renegadeinsurance.com*

- React + TypeScript with Salesforce-connected workflows; 70–100 inquiries/week at about 20% conversion.

Sarang — Travel discovery & trip-planning platform *saarang.com.np*

- Responsive frontend with interactive itineraries, built for low-bandwidth users at Young Innovations.

Government & Public Sector

Census Nepal 2021 — National census portal, Central Bureau of Statistics *censusnepal.cbs.gov.np*

- Public-facing, accessible visualization layer for national demographic data; built at Young Innovations.

MuAN — Internal admin dashboard, Municipal Association of Nepal *muannepal.org.np*

- Data visualization, user management and daily reporting for the umbrella body of 293 municipalities.

DGM Nepal — Internal admin dashboard, World Bank–funded grant program *dgmnepal.org*

- Operational reporting and user management for the Dedicated Grant Mechanism supporting Indigenous Peoples and local communities.

Open Source & Personal

a11y-beast — Open-source web accessibility scanner *github.com/bishojbk/a11y-beast*

- **125+ automated rules** mapped to WCAG 2.2, ADA, EAA, Section 508 and VPAT 2.5; Next.js, axe-core and Puppeteer on zero-cost infrastructure for free public access.

clairvoy — Cost optimizer for Claude Code *github.com/bishojbk/clairvoy*

- Cuts Claude Code spend by **20–40%** with data-driven `CLAUDE.md` rules generated from real usage patterns.

Nepal Tax Calculator — Bilingual salary tax calculator *github.com/bishojbk/nepal-tax-calc*

- Models Nepal's IRD tax rules end to end (SSF/CIT, retirement-cap crossover, single vs. couple filing, foreign income) with English/Nepali i18n; Jest-tested against real payslip data.

Skills

- **Frontend:** React, Next.js (App Router, Server Components, SSR), TypeScript, JavaScript, HTML, CSS
- **State Management & Forms:** TanStack Query (React Query), Zustand, React Hook Form, Zod
- **UI & Design Systems:** Tailwind CSS, shadcn/ui, Radix UI, Framer Motion, Ladle, Figma
- **Testing & Quality:** Vitest, Jest, Playwright, axe-core, Puppeteer, Biome
- **Tooling & Infrastructure:** Turborepo, pnpm, Git, GitHub Actions (CI/CD), Docker, Sentry, Datadog
- **Accessibility & Performance:** WCAG 2.2, ARIA, Core Web Vitals, code splitting, virtualization, i18n
- **Backend & Platform:** Node.js, REST APIs, Salesforce (Apex/LWC), Java
- **AI-assisted development:** Claude Code, Cursor, GitHub Copilot; repo-level `CLAUDE.md` conventions
- **Languages:** English (fluent), Nepali (native)

Education

Tribhuvan University, *Thapathali Campus, Kathmandu* *2020 – 2024*

Bachelor of Engineering (B.E.) in Electronics, Communication & Information Engineering